

Floyd Labs Website — Complete Handoff Documentation

Project: FloydTheWebsite

Owner: Douglas Talley (Floyd Labs / Legacy AI)

Domain: floydslabs.com (<https://floydslabs.com>)

GitHub: [LegacyAI-FloydsLabs/FloydTheWebsite](https://github.com/LegacyAI-FloydsLabs/FloydTheWebsite) (<https://github.com/LegacyAI-FloydsLabs/FloydTheWebsite>)

Last Updated: May 5, 2026

Commit: `c0d8ac1`

1. Architecture Overview

Stack

Layer	Technology	Notes
Framework	Next.js 14 (App Router)	Pages use Server Components by default, Client Components where interactivity needed
Language	TypeScript (strict)	<code>tsconfig.json</code> with path aliases (<code>@/</code> → project root)
Database	PostgreSQL 17 (Neon)	Prisma ORM, pooled connection via Neon
Auth	NextAuth.js v4	Credentials (email/password) + Google SSO, JWT strategy
Styling	Tailwind CSS 3 + custom CSS vars	Cyberpunk/neon theme, glassmorphism panels
UI Components	Radix UI + ShadCN primitives	35+ components in <code>components/ui/</code>
Animations	Framer Motion 10	Hero section, stats counters, page transitions
Hosting	Vercel (primary)	Auto-deploy on push to <code>main</code> branch
DB Hosting	Neon Postgres	Pooled endpoint, <code>?sslmode=require</code>
DNS/Domain	floydslabs.com	Configured via Vercel

Companion Sites

Site	URL	Purpose
Legacy AI	www.LegacyAI.space (https://www.LegacyAI.space)	Parent brand / ecosystem hub
GitHub Org	github.com/LegacyAI-FloydsLabs (https://github.com/LegacyAI-FloydsLabs)	Open source repos

2. Directory Structure

```

/
├── app/
│   ├── components/
│   │   ├── hero-section.tsx
│   │   ├── stats-section.tsx
│   │   ├── about/page.tsx
│   │   ├── admin/
│   │   │   ├── api-keys/page.tsx
│   │   │   ├── apps/
│   │   │   ├── blog/
│   │   │   ├── contacts/page.tsx
│   │   │   ├── login/page.tsx
│   │   │   ├── tools/page.tsx
│   │   │   ├── layout.tsx
│   │   │   └── page.tsx
│   │   ├── api/
│   │   │   ├── admin/
│   │   │   │   ├── auth/[...nextauth]/
│   │   │   │   ├── contact/route.ts
│   │   │   │   ├── mcp/
│   │   │   │   └── signup/route.ts
│   │   │   ├── api-docs/page.tsx
│   │   │   ├── apps/page.tsx
│   │   │   ├── blog/
│   │   │   │   ├── page.tsx
│   │   │   │   └── [slug]/page.tsx
│   │   │   ├── connect/page.tsx
│   │   │   ├── contact/
│   │   │   ├── globals.css
│   │   │   ├── layout.tsx
│   │   │   ├── not-found.tsx
│   │   │   ├── page.tsx
│   │   │   ├── privacy/page.tsx
│   │   │   ├── robots.ts
│   │   │   ├── sitemap.ts
│   │   │   └── terms/page.tsx
│   │   ├── components/
│   │   │   ├── footer.tsx
│   │   │   ├── navbar.tsx
│   │   │   ├── providers.tsx
│   │   │   ├── theme-provider.tsx
│   │   │   ├── theme-toggle.tsx
│   │   │   └── ui/
│   │   └── hooks/
│   │       └── use-toast.ts
│   ├── lib/
│   │   ├── auth.ts
│   │   ├── blog-data.ts
│   │   ├── db.ts
│   │   ├── mcp/
│   │   │   ├── auth.ts
│   │   │   ├── executor.ts
│   │   │   ├── servers.ts
│   │   │   ├── skills.ts
│   │   │   └── types.ts
│   │   ├── types.ts
│   │   └── utils.ts
│   ├── prisma/
│   │   ├── schema.prisma
│   │   └── public/
│   │       ├── Floydslabs.png
│   │       └── bg-graffiti.png
├── Next.js App Router pages
├── Homepage-specific components
├── Animated hero with framer-motion
├── Animated counters (73 tools, 3 servers, etc.)
├── About page team, philosophy, ecosystem
├── Protected admin panel
├── API key management
├── Application CRUD
├── Blog post CRUD
├── Contact submissions viewer
├── Admin login form
├── Skills management
├── Admin layout with sidebar
├── Admin dashboard
├── API routes
├── Admin CRUD endpoints
├── NextAuth catch-all
├── Contact form submission
├── MCP Server API (see §5)
├── User registration
├── Swagger UI for MCP API
├── Floyd Suite showcase
├── Blog listing + detail
├── Blog index (DB + static fallback)
├── Individual post (markdown rendered)
├── MCP connection guide
├── Contact page + form component
├── Theme: CSS vars, glass-panel, floyd-card
├── Root layout (meta, fonts, providers)
├── Custom 404
├── Homepage
├── Privacy policy
├── Dynamic robots.txt
├── Dynamic sitemap.xml
├── Terms of service
├── Shared components
├── Global footer (ecosystem links)
├── Global nav (7 links + mobile menu)
├── SessionProvider + ThemeProvider wrapper
├── next-themes provider
├── Dark/light toggle
├── 35+ ShadCN/Radix components
├── Toast notification hook
├── NextAuth config (Credentials + Google)
├── Static blog post data (fallback)
├── Prisma client singleton
├── MCP server library
├── JWT token generation/validation
├── Skill execution engine
├── 3 server definitions + tool mapping
├── Skill fetching from DB
├── TypeScript interfaces
├── App-wide types
├── cn() utility (clsx + tailwind-merge)
├── Database schema (11 models)
├── Logo
├── Background texture

```

og- image .png	# Open Graph social share image
scripts/	
populate-skills.ts	# Bulk skill import
safe-seed.ts	# Non-destructive seed
seed.ts	# Initial data seed
types/	
next -auth.d.ts	# NextAuth type augmentations
.env.example	# Environment variable template
.gitignore	
components.json	# ShadCN component config
next .config.js	# Next .js configuration
package.json	# Dependencies
postcss.config.js	# PostCSS (Tailwind)
tailwind.config.ts	# Tailwind theme extensions
tsconfig.json	# TypeScript config
vercel.json	# Vercel build config
yarn.lock	# Dependency lockfile

3. Database Schema

Models (11 total)

Model	Table	Purpose	Key Fields
session	sessions	MCP session tracking	session_id , expires_at , metadata
usage_tracking	usage_tracking	Tool execution metrics	tool_name , server , execution_time , success
skills_cache	skills_cache	73 AI skills — the core data	skill_name (PK), content (markdown), schema (JSON)
contact_submission	contact_submissions	Contact form entries	name , email , message , status
User	users	Admin/auth users	email , password (bcrypt), role , OAuth fields
Account	accounts	OAuth account links	provider , provider-AccountId , access_token
VerificationToken	verification_tokens	Email verification	identifier , token , expires
blog_post	blog_posts	Blog content	slug , title , content (markdown), published
application	applications	Floyd Suite apps	name , slug , status , features[]
api_key	api_keys	MCP API authentication	key (unique), tier , rate_limit , calls_today
mcp_call_log	mcp_call_logs	MCP execution audit trail	api_key_id , skill_name , duration_ms , success

Row Counts (as of May 5, 2026)

Table	Rows
skills_cache	73
blog_posts	5
users	4
applications	4
contact_submissions	3
accounts	1
Others	0 (empty, ready for use)

Connection String Format

```
postgresql://<user>:<password>@<host>/<database>?sslmode=require
```

- Uses Neon's **pooled** endpoint (port 5432 via `-pooler` hostname)
 - `?sslmode=require` is mandatory
 - Do NOT add `?channel_binding=disable` — Neon pooler doesn't need it
-

4. Environment Variables

Variable	Required	Description
<code>DATABASE_URL</code>	✓	Neon Postgres pooled connection string
<code>NEXTAUTH_SECRET</code>	✓	Random secret for JWT signing (<code>openssl rand -base64 32</code>)
<code>NEXTAUTH_URL</code>	✓ (prod)	<code>https://floydslabs.com</code> — Vercel sets this automatically
<code>NEXT_PUBLIC_SITE_URL</code>	Recommended	<code>https://floydslabs.com</code> — used in sitemap/robots
<code>GOOGLE_CLIENT_ID</code>	For Google SSO	Google OAuth 2.0 client ID
<code>GOOGLE_CLIENT_SECRET</code>	For Google SSO	Google OAuth 2.0 client secret
<code>ABACUSAI_API_KEY</code>	Optional	Legacy — was used for Abacus LLM APIs

Vercel Environment Setup

All vars must be set in Vercel Dashboard → Project Settings → Environment Variables for **Production**, **Preview**, and **Development** environments.

5. MCP Server System

The MCP (Model Context Protocol) API is the technical backbone of Floyd Labs — it exposes 73 AI skills as callable tools for LLMs.

Architecture

```

Client (LLM/Agent)
├── /api/mcp/auth → JWT Token
├── /api/mcp/servers → 3 servers listed
├── /api/mcp/servers/{name}/tools → tools for that server
├── /api/mcp/skills → all 73 skills
├── /api/mcp/skills/{name} → skill detail + schema
├── /api/mcp/skills/{name}/execute → run a skill
├── /api/mcp/health → system health
├── /api/mcp/metrics → usage stats
└── /api/mcp/api-docs-json → OpenAPI 3.0 spec

```

The 3 MCP Servers

Server	Tools	Domain
floyd-core	19 mapped	Dev ops, code analysis, build tools, git
ai-cognition	22 mapped	Pattern recognition, semantic analysis, knowledge synthesis
ai-orchestration	26 mapped	Multi-agent coordination, task management, swarm intelligence

Total mapped tools: 67 (explicitly assigned to servers)

Total skills in DB: 73 (includes 6 unassigned + Ghost Algorithms)

Ghost Algorithms: 10 (autonomous emergent behaviors, not server-bound)

Authentication Flow

1. POST `/api/mcp/auth` with `{ password: "<shared_password>" }`
2. Receive JWT token (24h expiry)
3. Include `Authorization: Bearer <token>` on all subsequent requests

Key Files

- `lib/mcp/servers.ts` — Server definitions + tool→server mapping (3 `Set<string>` collections)
- `lib/mcp/auth.ts` — JWT generation/validation using `jsonwebtoken`
- `lib/mcp/executor.ts` — Skill execution engine (fetches from DB, runs logic)
- `lib/mcp/skills.ts` — DB queries for skills
- `lib/mcp/types.ts` — TypeScript interfaces
- `app/api/mcp/api-docs-json/route.ts` — Full OpenAPI 3.0 spec generator

Tool Name Convention

- **DB stores:** hyphenated (`build-error-correlator`)
- **Server mapping uses:** underscored (`build_error_correlator`)

- `getServerForSkill()` in `servers.ts` normalizes by replacing `-` with `_`
- 6 tools in the server mapping don't have exact DB matches (cosmetic gap, not functional)

6. Authentication System

Providers

1. **Credentials** — email + bcrypt-hashed password
2. **Google SSO** — OAuth 2.0 via `next-auth/providers/google`
 - `allowDangerousEmailAccountLinking: true` — links Google to existing email accounts

Session Strategy

- JWT-based (not database sessions)
- Token includes: `id`, `role`, `email`, `name`
- Session callback exposes these to client via `useSession()`

Protected Routes

- `/admin/*` — requires authenticated session with admin role
- `/api/admin/*` — API routes check `getSession(authOptions)`
- `/api/mcp/*` — uses separate JWT auth (MCP shared password, not NextAuth)

Admin Users

Admin panel accessible at `/admin/login`. Users in `users` table with `role: "admin"`.

7. Theming & Design System

Visual Identity

- **Theme:** Neon cyberpunk with glassmorphism
- **Colors:** Cyan (`#00e5ff`) primary accent, Magenta/Purple (`#9c27b0`, `#ce93d8`) secondary, Dark backgrounds (`#0a0514`)
- **Font:** JetBrains Mono (monospace), Inter (body)
- **Background:** `bg-graffiti.png` texture visible through glass panels

CSS Architecture

- `globals.css` defines all CSS variables under `:root` / `.dark` selectors
- Two key utility classes:
 - `.floyd-card` — 0.95 opacity dark bg, blur(16px), purple border glow
 - `.glass-panel` — 0.95 opacity, blur(20px), 2px purple border, drop shadow
- Both ensure WCAG AA text contrast against the graffiti background
- Dark mode is default; light mode supported but secondary

Component Library

35+ Radix/ShadCN components in `components/ui/` — all themed to match the cyberpunk aesthetic via Tailwind config extending CSS variables.

8. Blog System

Dual Source

1. **Database** (`blog_posts` table) — primary, checked first
2. **Static data** (`lib/blog-data.ts`) — fallback if DB empty/unreachable

Content Format

- Markdown stored in `content` field
- Rendered client-side using `marked` library
- Tags stored as `String[]` in Postgres

Current Posts (5)

1. The Garage Chronicles: Origins Edition
2. The Subscription Hater's Manifesto
3. The Suite
4. The Gospel According to Nick Beard
5. The Gas Station Manifesto

9. SEO & Metadata

- `metadataBase` : `https://floydslabs.com`
- Open Graph image: `/og-image.png`
- Dynamic `robots.ts` : allows all, disallows `/api/`
- Dynamic `sitemap.ts` : all static pages + blog slugs, includes `/connect` (0.8) and `/api-docs` (0.7)
- Each page exports its own `Metadata` object

10. Deployment (Vercel)

Setup

- **Git Integration:** Auto-deploy on push to `main` branch
- **Build Command:** `prisma generate && next build` (set in `vercel.json`)
- **Framework:** Next.js (auto-detected)
- **Node.js:** 18.x (Vercel default)

Critical: `next.config.js` for Vercel

The Vercel-clean config must be:

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  eslint: { ignoreDuringBuilds: true },
  typescript: { ignoreBuildErrors: false },
  images: { unoptimized: true },
};
module.exports = nextConfig;
```

⚠ Do NOT include `outputFileTracingRoot`, `distDir`, `output`, or custom webpack config — these are Abacus-specific and cause Vercel's `path0/path0` build error.

Vercel Project Info

- **Team:** `team_0nEd9CwoThv78d03w9BfySZd`
- **Project:** `prj_onnNfGYigKSZg3RbenrYZin86bZR`

11. Known Issues & Gotchas

Issue	Details	Workaround
<code>next.config.js</code> divergence	Abacus VM version has <code>outputFileTracingRoot</code> that breaks Vercel	Use the clean config (\$10) for Vercel
Tool name mismatch	6 tools in server mapping use underscored names not matching DB hyphenated names	<code>getServerForSkill()</code> normalizes, but count shows 13/19 for floyd-core
Signup test endpoint	Automated tests hit <code>/api/auth/signup</code> but real endpoint is <code>/api/signup</code>	Pre-existing, not a bug
<code>prisma/schema.prisma</code> output path	Has Abacus-specific path in <code>output</code> field	Remove <code>output</code> and <code>binaryTargets</code> for Vercel (Prisma auto-detects)
Blog static fallback	If DB is unreachable, falls back to <code>lib/blog-data.ts</code> which may be stale	Keep both in sync
Google SSO cookies	NextAuth v4 requires explicit cookie config for OAuth state persistence	Already configured in <code>lib/auth.ts</code>

12. Vercel Migration Checklist

When fully migrating from Abacus to Vercel:

- [x] Push code to GitHub (clean structure, no `nextjs_space/` nesting)
- [x] Import to Neon Postgres (all 11 tables, 73 skills)
- [x] Set environment variables in Vercel dashboard
- [] Fix `next.config.js` — remove Abacus-specific options
- [] Fix `prisma/schema.prisma` — remove `output` and `binaryTargets`

- [] Verify Google SSO callback URL in Google Console → `https://floydslabs.com/api/auth/callback/google`
 - [] Verify custom domain DNS in Vercel
 - [] Test all routes post-deploy
 - [] Set up Vercel Analytics (optional)
-

13. Quick Reference Commands

```
# Local development
yarn install
yarn prisma generate
yarn dev                # http://localhost:3000

# Database
yarn prisma db push      # Apply schema changes (careful!)
yarn prisma studio       # Visual DB browser
yarn prisma generate     # Regenerate Prisma client

# Seeding
npx tsx scripts/seed.ts   # Initial seed
npx tsx scripts/safe-seed.ts # Non-destructive upsert
npx tsx scripts/populate-skills.ts # Bulk skill import

# Type checking
yarn tsc --noEmit

# Production build
yarn build
yarn start
```

14. Content Accuracy Reference

These numbers must be consistent across all pages:

Metric	Value	Where it appears
Total Skills	73	Homepage, About, Tools, Connect, Layout metadata
MCP Servers	3	About, Tools, Connect, API Docs
Server Names	floyd-core, ai-cognition, ai-orchestration	Connect, Tools, API Docs
Mapped Tools	67	Connect, API Docs header
Ghost Algorithms	10	About (stats), Tools (header text)
Subscriptions	\$0	Homepage, About, everywhere
Cost	\$0/month	Homepage stats, Tools

This document is the single source of truth for the Floyd Labs website. Keep it updated with every significant change.